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1 "C:\Program Files\Python313\Scripts\uv.exe" run "C:/Users/Prajwal Anand/Desktop/sem7_project/.venv/Scripts/python.exe" "C:\Users\
   Prajwal Anand\Desktop\sem7_project\t3\t3.py"
2 C:\Users\Prajwal Anand\Desktop\sem7_project\t3\t3.py:133: DeprecationWarning: Conversion of an array with ndim > 0 to a scalar is
   deprecated, and will error in future. Ensure you extract a single element from your array before performing this operation. (
   Deprecated NumPy 1.25.)
3 print(" %7.5f %7.1f %7.3f %7.1f %7.1f %7.1f %7.3f %7.5f" %summary_data, file=file, flush=True)
4 RUN MODEL FINISHED
5
6 ===== DESIGN SCALING FACTORS =====
7
8 Corrected AirFlow (Wc): [2.85722494]
9 Corrected Speed (Nc): [2997.00315988]
10 Scale Wc (s_Wc): [0.09524083]
11 Scale PR (s_PR): [1.666666667]
12 Scale Eff (s_eff): [0.97532315]
13 Scale Nc (s_Nc): [2997.00315988]
14
15 ===== TURBINE SCALING =====
16
17 Corrected Flow: [1.19432594]
18 Scale Flow: [0.00796759]
19 Scale PR: [0.36471565]
20 Scale Eff: [0.92712376]
21 Scale Speed: [0.63960215]
22
23 ===== NOZZLE =====
24
25 Throat Area: [5.80444684]
26 Exit Area: [6.25297835]
27
28 ===== ENGINE SIZE =====
29
30 Mass Flow: [1.30319747]
31 Net Thrust: [0.89324285]
32 Fuel Flow: [0.02258047]
33
34
35
36 -----
37 POINT: DESIGN
38 -----
39 PERFORMANCE CHARACTERISTICS
```

40	Mach	Alt	W	Fn	Fg	Fram	OPR	TSFC				
41	0.10000	0.0	1.303	0.9	0.9	0.0	8.000	0.00003				
42	-----											
43	FLOW STATIONS											
44	-----											
45	-----											
46	Flow Station			tot:P	tot:T	tot:h	tot:S	stat:P	stat:W	stat:MN	stat:V	
47	stat:area	-----										
48	DESIGN.fc.FL_0			14.792	519.708	-5.930	1.631	14.696	2.205	0.100	111.650	
	37.050											
49	DESIGN.inlet.FL_0			14.792	519.708	-5.930	1.631	11.674	2.873	0.600	647.738	
	9.815											
50	DESIGN.comp.FL_0			118.337	1017.885	115.374	1.652	118.302	2.873	0.020	31.051	
	42.432											
51	DESIGN.burner.FL_0			114.787	2200.000	113.409	1.880	114.759	2.923	0.020	44.450	
	67.233											
52	DESIGN.turb.FL_0			40.653	1782.584	-5.829	1.891	36.584	2.923	0.400	795.505	
	9.306											
53	DESIGN.nozz.FL_0			40.653	1782.584	-5.829	1.891	14.696	2.923	1.312	2343.648	
	6.253											
54	-----											
55	-----											
56	COMPRESSOR PROPERTIES											
57	-----											
58	Compressor		Wc	Pr	eta_a	eta_p	Nc	pwr	RlineMap	NcMap	PRmap	WcMap
	alphaMap	SMN	SMW	effMap								
59	-----											
60	DESIGN.comp		2.857	8.000	0.830	0.871	2997.003	-493.096	2.000	1.000	5.200	30.000
	0.000	20.000	19.189	0.851								
61	-----											
62	-----											
63	BURNER PROPERTIES											

64	-----										
65	Burner		dPqP	TtOut	Wfuel	FAR					
66	DESIGN.burner		0.0300	2200.00	0.0498	0.01733					
67	-----										
--											
68	TURBINE PROPERTIES										
69	-----										
--											
70	Turbine		Wp	PR	eff_a	eff_p	Np	pwr	NpMap	PRmap	
	alphaMap										
71	DESIGN.turb		1.194	2.824	0.860	0.844	63.960	493.096	100.000	6.000	1.
	000										
72	-----										
73	NOZZLE PROPERTIES										
74	-----										
75	Nozzle		PR	Cv	Cfg	Ath	MNth	MNout	V	Fg	
76	DESIGN.nozz		2.766	0.990	N/A	5.804	1.000	1.312	2343.648	210.779	
77	-----										
78	SHAFT PROPERTIES										
79	-----										
80	Shaft		Nmech		trqin	trgout	pwrin	pwrout			
81	DESIGN.shaft		3000.000		863.264	-863.264	493.095	-493.095			
82											
83											
84											
85	-----										
86	POINT: 0D0										
87	-----										
88	PERFORMANCE CHARACTERISTICS										
89	Mach	Alt	W	Fn	Fg	Fram	OPR	TSFC			
90	0.10000	0.0	1.303	0.9	0.9	0.0	8.000	0.00003			
91	-----										
--											
92	FLOW STATIONS										
93	-----										
--											
94	-----										
--											
95	Flow Station		tot:P	tot:T	tot:h	tot:S	stat:W	stat:MN	stat:V		
	stat:area										
96	-----										
--											

97	0D0.fc.FL_0 37.050		14.792	519.708	-5.930	1.631	14.696	2.205	0.100	111.650		
98	0D0.inlet.FL_0 9.815		14.792	519.708	-5.930	1.631	11.674	2.873	0.600	647.738		
99	0D0.comp.FL_0 42.432		118.337	1017.885	115.374	1.652	118.302	2.873	0.020	31.051		
100	0D0.burner.FL_0 67.233		114.787	2200.000	113.409	1.880	114.759	2.923	0.020	44.450		
101	0D0.turb.FL_0 9.306		40.653	1782.584	-5.829	1.891	36.584	2.923	0.400	795.505		
102	0D0.nozz.FL_0 6.253		40.653	1782.584	-5.829	1.891	14.696	2.923	1.312	2343.648		
103	-----											
104	-----											
105	COMPRESSOR PROPERTIES											
106	-----											
107	Compressor alphaMap		Wc SMW	Pr effMap	eta_a	eta_p	Nc	pwr	RlineMap	NcMap	PRmap	WcMap
108	-----											
109	0D0.comp 0.000		2.857 20.000	8.000 0.851	0.871	2997.003	-493.096	2.000	1.000	5.200	30.000	
110	-----											
111	-----											
112	BURNER PROPERTIES											
113	-----											
114	Burner		dPqP	TtOut	Wfuel	FAR						
115	0D0.burner		0.0300	2200.00	0.0498	0.01733						
116	-----											
117	TURBINE PROPERTIES											
118	-----											
119	Turbine alphaMap		Wp	PR	eff_a	eff_p	Np	pwr	NpMap	PRmap		
120	0D0.turb 000		1.194	2.824	0.860	0.844	63.960	493.096	100.000	6.000	1.	
121	-----											

NOZZLE PROPERTIES

Nozzle		PR	Cv	Cfg	Ath	MNth	MNout	V	Fg
0D0.nozz		2.766	0.990	N/A	5.804	1.000	1.312	2343.648	210.779

SHAFT PROPERTIES

Shaft		Nmech	trqin	trqout	pwrin	pwrout
0D0.shaft		3000.000	863.264	-863.264	493.095	-493.095

POINT: OD1

PERFORMANCE CHARACTERISTICS

Mach	Alt	W	Fn	Fg	Fram	OPR	TSFC
0.20000	1524.0	1.143	0.8	0.8	0.1	8.290	0.00003

FLOW STATIONS

Flow Station		tot:P	tot:T	tot:h	tot:S	stat:P	stat:W	stat:MN	stat:V
stat:area									

0D1.fc.FL_0		12.548	504.848	-9.495	1.635	12.228	2.205	0.200	219.447
21.906									
0D1.inlet.FL_0		12.548	504.848	-9.495	1.635	3.530	2.519	1.475	1355.950
9.815									
0D1.comp.FL_0		104.028	1005.677	112.333	1.658	103.999	2.519	0.020	30.604
42.432									
0D1.burner.FL_0		100.907	2200.000	110.402	1.889	100.882	2.563	0.020	44.347
67.233									
0D1.turb.FL_0		35.623	1780.923	-9.331	1.900	32.127	2.563	0.399	793.819
9.306									
0D1.nozz.FL_0		35.623	1780.923	-9.331	1.900	12.228	2.563	1.352	2397.219
6.350									

